MATH4528: Exam Prep

Martingales

Question 3

Subsets of F or $\mathcal{F}_\Omega$ = 2^n where n is the number of possible events in Ω.

Subsets of $\mathcal{F}_S$ where S is a Random Variable, $S(\omega) : \Omega \to \mathbb{R}$. Here $\mathcal{F}_S = \{S^{-1}(B) : B \in \mathbb{B}(\mathbb{R})\}$.

A Random Variable X, is $\mathcal{F}_S$ measurable if $\forall B, X^{-1}(B) \in \mathcal{F}_S$.

Finding distribution of $Z = E[X|\mathcal{F}_S]$:
1. find the most finite division of Ω.

$$\Omega = A \cup B \cup C, A \cap B \cap C = \emptyset$$

2. determine the form of Z as constant across the subdivisions.

$$Z = \begin{cases} z_1 & \text{if} \quad A \\ z_2 & \text{if} \quad B \\ z_3 & \text{if} \quad C \end{cases}$$

3. calculate each constant.

$$\forall A \in \mathcal{F}_S, \int_A X dP = \int_A Z dP$$

$$\int_A X dP = E[X 1_A] = z_a P(A) = \int_A Z dP$$

$$z_a = \frac{E[X 1_A]}{P(A)}$$

This works because since Z is $\mathcal{F}_S$-measurable, then it must be constant under $\mathcal{F}_S$.

Question 4

For two sigma fields such that $\mathcal{F} \subset \mathcal{G}$,

$$E[E[Y|\mathcal{G}]E[Y|\mathcal{F}]] = E[(E[Y|\mathcal{F}])^2]$$

we can apply the tower property $E[Z] = E[E[Z|F]]$.

$$E[E[Y|\mathcal{G}]E[Y|\mathcal{F}]] = E\{E[E[Y|\mathcal{G}]E[Y|\mathcal{F}] \mid \mathcal{F}]\}$$

here the smaller sigma field applies,

$$E[E[Y|\mathcal{G}]E[Y|\mathcal{F}]] = E\{E[Y|\mathcal{F}]E[E[Y|\mathcal{G}]|\mathcal{F}]\}$$

$$E[E[Y|\mathcal{G}]E[Y|\mathcal{F}]] = E\{E[Y|\mathcal{F}]E[Y|\mathcal{F}]\}$$

$$E[E[Y|\mathcal{G}]E[Y|\mathcal{F}]] = E\{(E[Y|\mathcal{F}])^2\}$$

Doob's Inequality:

$$P\left(\max_{m \le n} X_m \ge \lambda\right) = P\left(\max_{m \le n} X_m + c \ge \lambda + c\right) \le P\left(\max_{m \le n} (X_m + c)^2 \ge (\lambda + c)^2\right)$$

$$P\left(\max_{m \le n} (X_m + c)^2 \ge (\lambda + c)^2\right) \le \frac{E\left[(X_n + c)^2\right]}{(\lambda + c)^2}$$

Optimise for c by finding the derivative equal to zero

$$\frac{\delta}{\delta c} \frac{E[X_n^2] + c^2}{(\lambda + c)^2} = 0 \to c = \frac{E[X_n^2]}{\lambda}$$

Question 5

Sequence X_n is a martingale, if $E[X_{n+1} | \mathcal{F}_n] = X_n$.
Proof uses the fact that X_n is $\mathcal{F}_n$-integrable and that anything $n + 1$ is independent of $\mathcal{F}_n$.

For a stopping time $N = \inf\{n : S_n = 0\}$,

$$\{S_n \le 0\} \subset \{N \le n\}$$

which by taking compliments imples

$$\{N > n\} \subseteq \{S_n > 0\}$$

For a stopping time $N = \inf\{S_n = 0\}$, we can say that

$$E[N] \le \sum_{n=0}^{\infty} P(N > n)$$

If $X_{N \wedge n}$ is uniformly integrable and $N = \inf\{S_n = 0\}$ is a stopping time, then

$$E[X_N] = E[X_0]$$

Question 6

For a Brownian motion process $B(s)$, if $0 \le s \le t$, $E[B(s)B(t)] = s$.

If $\{B(s) : 0 \le s \le t\}$ is uniformly continuous, then it has a Riemann sum:

$$\int_0^t B(s) ds = \lim_{n \to \infty} \frac{t}{n} \sum_{k=1}^{n} B\left(\frac{t}{n} k\right)$$

$$X_n := \frac{t}{n} \sum_{k=1}^{n} B\left(\frac{t}{n} k\right)$$

which implies that X_n is Gaussian, but NOT i.i.d, since each $B\left(\frac{t}{n}k\right)$ is dependant, but we can split this up into independent increments.

$$X_n = \frac{t}{n}\sum_{k=1}^{n}\sum_{j=1}^{k} B\left(\frac{t}{n}j\right) - B\left(\frac{t}{n}(j-1)\right)$$

which we can simplify by counting the number of each increment in a grid

$$X_n = \frac{t}{n}\sum_{k=1}^{n}(n-k+1)\left[B\left(\frac{t}{n}k\right) - B\left(\frac{t}{n}(k-1)\right)\right]$$

which IS a sum of independant Gaussian random variables, so we can find mean and variance as expected. Follows from the linear combination of i.i.d Normals since,

$$B\left(\frac{t}{n}k\right) - B\left(\frac{t}{n}(k-1)\right) \sim N\left(0, \frac{t}{n}\right)$$

Probability

Question 1

Borel-Cantelli Lemmas, for a sequence of events $\{A_i\}_{i\geq 0}$,
First Lemma: if $\sum_{k=0}^{\infty} P(A_n) < \infty$, then $P\{A_n \text{ infinitely often}\} = 0$
Second Lemma: if $\sum_{k=0}^{\infty} P(A_n) = \infty$, then $P\{A_n \text{ infinitely often}\} = 1$

Infinitely Often and Eventually: $P\{A_n \text{ infinitely often}\} = 1 - P\{A_n \text{ eventually}\}$
If $X_{n,m}$ is a triangular array, $\bar{X}_n := \frac{1}{n}\sum_{m=1}^{n} X_{n,m}$
The following are equivalent:

$$P(A_n \text{ infinitely often}) = 0$$

$$P(A_n \text{ eventually}) = 1$$

$$P(\limsup A_n) = 1$$

Question 2

A triangular array $X_{n,k}$ has the condition of Infinite Smallness if for $\varepsilon > 0$ and for $n \to \infty$

$$\max_{k=1,\ldots,n} P\left(|X_{n,k}| \geq \varepsilon\right) \to 0$$

Where it is sufficient to consider the behaviour of $P\left(|X_{n,k}| \geq \varepsilon\right) \forall k$ as $n \to \infty$
For a discrete period $\{1, \ldots, n\}$, we can split these up into intervals from $\{1, \ldots, n-1\}$ and $\{2, \ldots, n\}$ forming the Upper and Lower Riemann Sums, based on whether $f(1)$ or $f(n)$ is larger.

$$\text{Lower Riemann Sum} \leq \int f(x)dx \leq \text{Upper Riemann Sum}$$

Lévy spectral function